Decoding Visual Stimuli from Neural Population Activity

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1 Introduction

Understanding how neural populations represent visual stimuli is a central question in computational neuroscience. In this project, using the MICrONs dataset, which provides access to neural activity across multiple areas of the visual hierarchy[1], we study how stimulus information is represented at different processing stages. Using only functional neural activity, we first test if the broad stimulus type can be classified from the visual cortical activity. Second, we ask whether the specific clip identity can be recovered from the neural responses. To address these questions, we compare static decoding models, based on population snapshots, with temporal models that use short sequences of neural activity, allowing us to evaluate whether temporal dynamics improve stimulus decoding. Finally, we perform this analysis both within primary visual cortex (V1) and across multiple visual areas (V1, LM, AL, RL), in order to compare how stimulus information is distributed along the visual hierarchy.

2 Methodology

2.1 Dataset and Preprocessing

The HDF5 MICrONs file was inspected to identify sessions, stimulus categories, neural responses, and brain-area labels. We selected session `4_7`, which contains 464 trials from three different stimulus categories [Table 1]: natural movie clips (*Clip*), stylized Monet stimuli (*Monet2*), and artificial psychedelic patterns (*Trippy*). In Figure 1 is reported one example frame for each category. The session includes 7493 neurons in total, across the four different visual areas [Table 2].

Trial frames were grouped using their video hash labels, where each hash identifies a unique stimulus. We verified that neuron counts were consistent across trials and stimulus categories, ensuring that comparisons across responses referred to the same neural populations. Because the number of recorded neurons differed substantially across areas, the comparisons across brain regions were performed by matching population size to the smallest one (AL, 387 neurons). V1, LM, and RL were randomly subsampled without replacement across 30 seeds averaging the results, while AL used all neurons.

2.2 Decoding Tasks

We started our analysis with the stimulus-type classification among the three categories. To avoid data leakage, repeated presentations were deduplicated at the hash level so that the same stimulus did not appear in both training and testing [Table 3]. To ensure a balanced dataset, we selected 20 unique stimuli per category, randomly subsampling the *Clip* class. This led to a baseline accuracy of 1/3.

Then, exact clip-identity decoding was formulated as a 96-class classification task. From the initial 240 unique clips, those shown only once were excluded because they could not support an independent train/test split. For the ninety clips that are shown twice, one repetition is used for training and one for testing, while for the six clips shown ten times, five repetitions are used for training and the other half for testing. This tested whether neural representations were stable across independent stimulus presentations.

2.3 Static and Temporal Models

Static decoding used Random Forest classifiers [2] and L2-regularized Logistic Regression. For stimulus-type decoding, each trial was represented by 11 population activity snapshots uniformly spaced, ignoring temporal order. For clip-identity decoding, all available timepoints from the retained clip repetitions were used as static population vectors.

Temporal decoding, instead, was performed using a gated recurrent unit (GRU), a recurrent architecture designed to model sequential dependencies through gated hidden states [3]. For the temporal models, neural responses were divided into non-overlapping 10-timepoint windows, allowing the decoder to incorporate short-term temporal structure rather than relying only on instantaneous neural snapshots. To reduce dimensionality and improve computational efficiency, the high-dimensional neural activity at each timepoint was first projected into a 64-dimensional latent space before being passed to the recurrent layer.

2.4 Evaluation and Interpretability

Models were evaluated analyzing level of accuracy. Performance was compared against chance levels of 0.3333 for stimulus-type classification and 0.0104 for exact clip-identity decoding.

For temporal models, gradient-based saliency was computed to identify which neurons and timepoints contributed most to decoding. This enabled us to assess whether stimulus information was concentrated in a small subset of neurons or distributed across the population, and whether decoding relied more strongly on early or late activity within the response window.

3 Results

3.1 Stimulus-Type Decoding

Stimulus type was decoded well above chance in all areas. With all 5312 V1 neurons, Logistic Regression

reached 0.8409 accuracy, Random Forest 0.7879, and the temporal GRU lifted performance further to 0.9052. Stimulus category is therefore strongly represented in the V1 population and is largely linearly separable when the full neural population is available. The confusion matrices show that *Trippy* is the most accurately classified category in both static classifiers, but the ranking reverses with the GRU, where *Monet2* becomes the easiest and *Trippy* the hardest, suggesting that static and temporal models exploit different features of the same stimuli (Figures 2, 3).

Under matched 387-neuron populations, Random Forest, which consistently outperformed Logistic Regression, reached 0.7955 in AL, 0.7528 in LM, 0.7487 in RL and 0.7149 in V1 with AL leading partly because it uses its full population without subsampling (Figure 4). The temporal GRU reached 0.8100 in V1, 0.7840 in LM, 0.7903 in AL and 0.7988 in RL (Figure 5). Majority-vote confusion matrices identify the same pattern as in the V1 case (Figures 6, 7, 8).

Compared with the best static model at the same neuron count, temporal modelling improved accuracy the most in V1, with smaller gains in RL and LM, but a slight loss in AL suggesting that category information is present not only in static firing patterns, but depending on the area, also in the temporal evolution of population activity.

3.2 Clip-Identity Decoding

Exact clip-identity decoding was substantially harder, with lower overall accuracy. In the full V1 model, static Logistic Regression and temporal GRU performed similarly, reaching 0.2364 and 0.2500, respectively, while Random Forest stayed lower at 0.1511, all clearly above the 0.0104 chance baseline (Figures 9, 10).

Under area-matched 387-neuron populations, LM led the cross-area ranking for both decoders, reaching 0.1616 with the best static model (Random Forest) and 0.1599 with the temporal GRU. The other areas followed with slightly lower levels of accuracy in both static and temporal decoding (see Figures 11 and 12). Overall, these results show that identifying the exact clip requires a more specific neural representation.

A major finding was the effect of repetition: clips shown ten times were decoded much more accurately than clips shown only twice, as shown in Figures 11 and 12. This indicates that repeated presentations lead to more reliable responses to the same stimulus, while rare clips remain strongly affected by trial-to-trial neural variability and limited training data.

4 Discussion

Three main points emerge from the previous results. First, broad stimulus category is recovered well above chance from every recorded area. We do not interpret this as evidence of semantic representation in primary visual cortex; rather, the three stimulus groups differ in low- and mid-level visual statistics (texture, contrast, spatial structure, motion) that are strongly reflected throughout the visual hierarchy [5]. The activity traces of the most informative neurons, identified via feature importance and saliency analyses, show that category information is distributed across heterogeneous, partially selective neurons rather than concentrated in a few highly selective cells (Figure 13). In the temporal case, saliency tends to peak later in the response window, suggesting that decoding benefits from accumulated population dynamics rather than purely early activity (Figure 14).

Second, individual natural clips evoke reproducible population signatures; specifically, the 96-class GRU trained on one set of repetitions generalizes to unseen trials, as it was previously documented in mouse V1 [4]. LM leading both static and temporal clip-identity decoders across the four brain areas, despite having far fewer recorded neurons than V1, suggests that intermediate visual areas may carry more clip-specific information per neuron, possibly because they integrate local features typical of V1 into more global spatiotemporal representations. The strong repetition effect further indicates that single-trial natural movie responses are dominated by trial-to-trial variability that no model can recover from a single example.

Finally, temporal modeling consistently improved decoding performance across conditions. In the full V1 population, the temporal GRU outperformed both static Random Forest and Logistic Regression for stimulus type, indicating that temporal dynamics provide additional information beyond static population snapshots (Figure 15). This advantage persists at the brain area level (Figure 16). Together, these results suggest that temporal integration is not only beneficial under constrained sampling conditions, but represents a generally informative dimension of neural population activity.

5 Conclusion

The main limitation is that decoding does not prove mechanism. A classifier can reveal that information is present in neural activity, but it does not prove that the brain uses the same variables or computations. Furthermore, the analysis used one session, so conclusions may not generalize across animals or recording conditions.

Future work should aim at testing the robustness and generality of these findings across different recording sessions and animals, in order to assess the stability of the neural representations over time and conditions. In addition, an important direction is to investigate the role of temporal integration more systematically, by quantifying how different timescales of neural activity contribute to stimulus representation.

Overall, the results support a distributed and temporally structured view of visual coding. Static population activity already contains substantial information about stimulus identity, but temporal dynamics provide additional information, with the largest gain in primary visual cortex and a magnitude that varies across areas and tasks.

References

[1] MICrONs Consortium. *Functional connectomics spanning multiple areas of mouse visual cortex*. Nature **640**, 435–447, 2025. doi: 10.1038/s41586-025-08790-w.

[2] Breiman, L. *Random Forests*. Machine Learning **45**, 5–32, 2001. doi: 10.1023/A:1010933404324.

[3] Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., and Bengio, Y. *Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation*. Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing, 1724–1734, 2014. doi: 10.3115/v1/D14-1179.

[4] Xia, J., Marks, T. D., Goard, M. J., and Witten, I. B. *Stable representation of a naturalistic movie emerges from episodic activity with gain variability*. Nature Communications **12**, 5170, 2021. doi: 10.1038/s41467-021-25437-2.

[5] Bolaños, F., et al. *Efficient coding of natural images in the mouse visual cortex*. Nature Communications **15**, 1787, 2024. doi: 10.1038/s41467-024-45919-3.

Appendix

A Preprocessing

Stimulus type	Number of trials
Clip	384
Monet2	40
Trippy	40

Table 1: Stimulus composition of session 4_7.

Visual area	Number of neurons
V1	5312
LM	983
AL	387
RL	811

Table 2: Number of recorded neurons per visual area in session 4_7.

Stimulus type	Number of unique hashes
Clip	240
Monet2	20
Trippy	20

Table 3: Unique hashes per category (after deduplication)

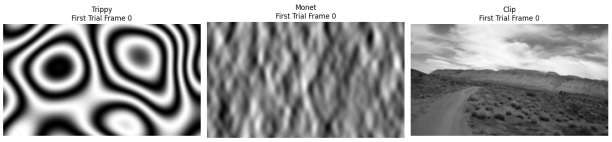


Figure 1: Frames of the three stimulus categories: *Trippy*, *Monet2*, and *Clip*.

B Stimulus-Type Analysis

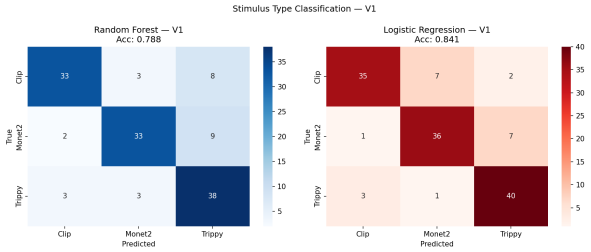


Figure 2: Confusion matrices for static V1 analysis.

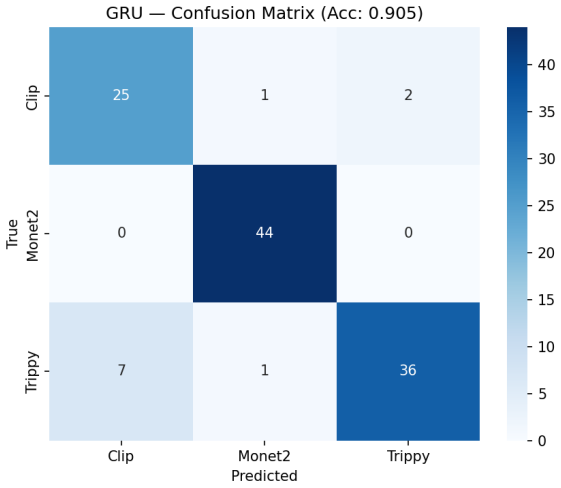


Figure 3: V1 GRU confusion matrix.

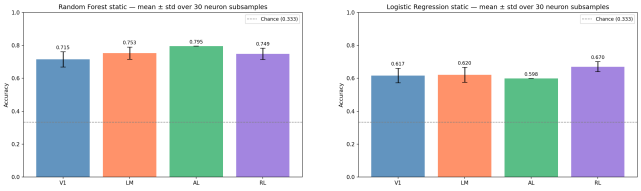


Figure 4: Static analysis across four brain areas.

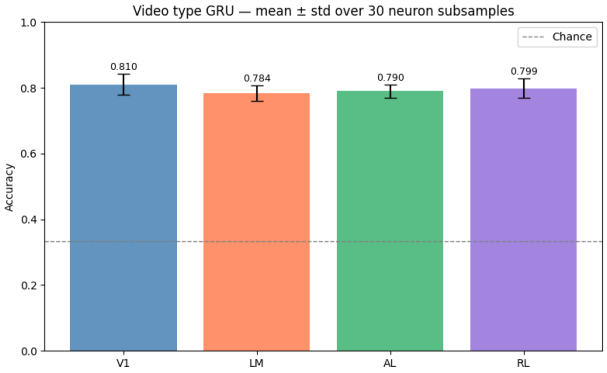


Figure 5: Temporal GRU performance across matched 387-neuron populations.

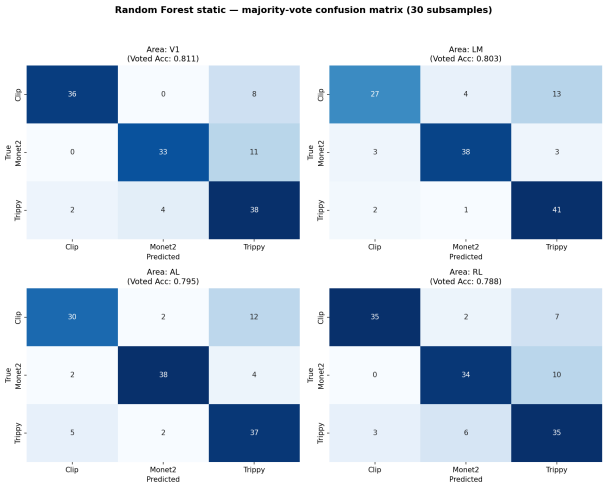


Figure 6: Majority-vote confusion matrices for static Random Forest.

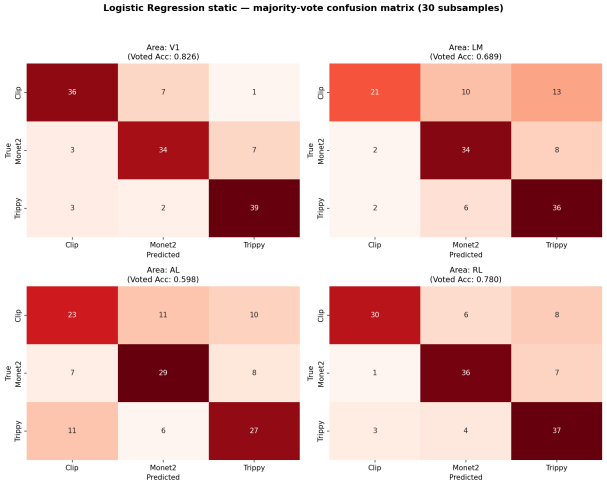


Figure 7: Majority-vote confusion matrices for static Logistic Regression.

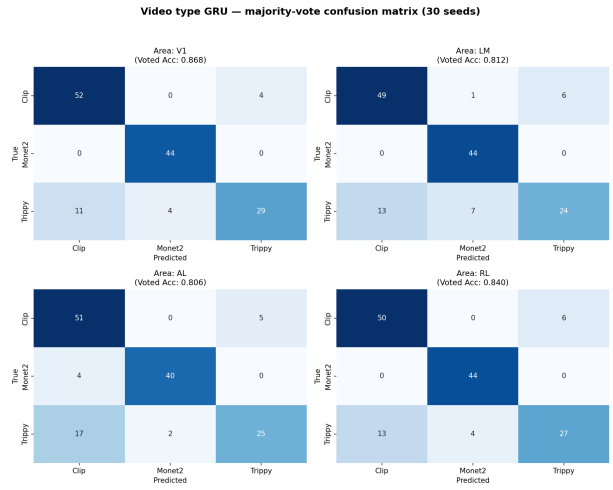


Figure 8: Majority-vote confusion matrices for temporal GRU.

C Clip-Identity Analysis



Figure 9: Static clip-identity decoding in V1. Left: Random Forest. Right: Logistic Regression.

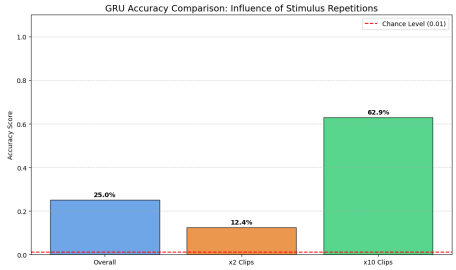


Figure 10: Temporal GRU clip-identity decoding in V1.

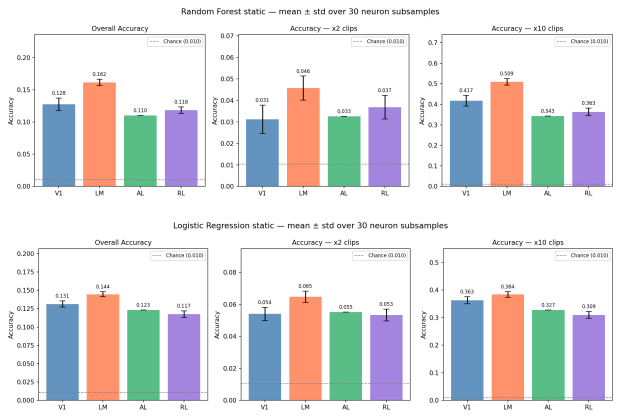


Figure 11: Random subsampling to 387 neurons. Top: Random Forest. Bottom: Logistic regression

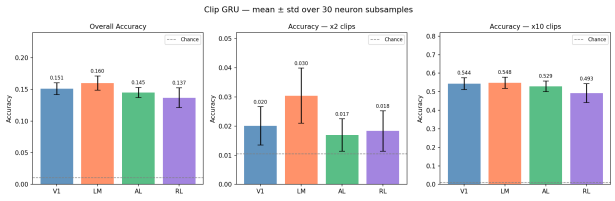


Figure 12: Random subsampling to 387 neurons. GRU decoding across areas.

D Final Analysis and Comparison

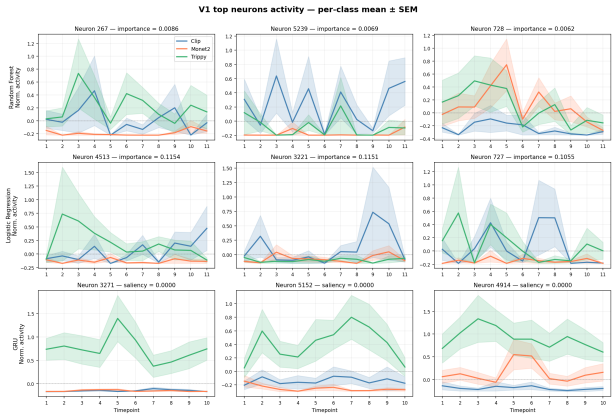


Figure 13: Activity traces of the most informative V1 neurons for each stimulus category.

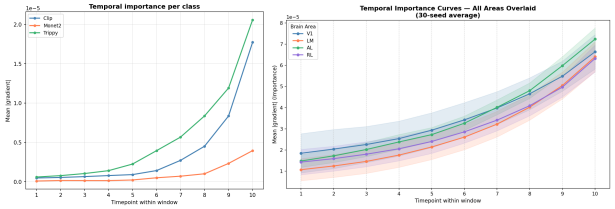


Figure 14: Temporal importance across timepoints within the analysis window. Left: class-specific saliency profiles. Right: temporal importance averaged across brain areas.

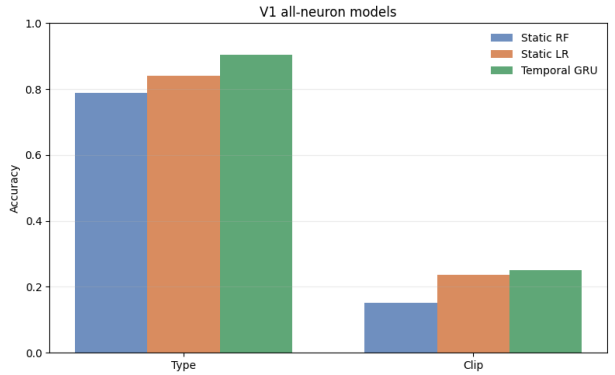


Figure 15: V1 all-neuron models comparison.

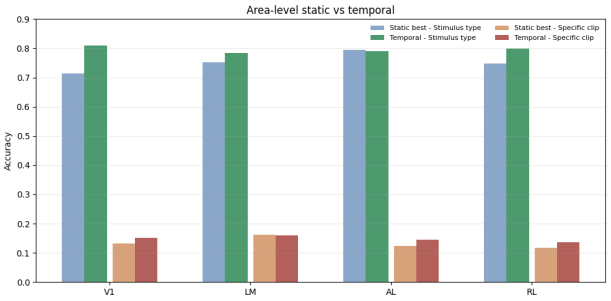


Figure 16: Final area-level comparison.